Annex IV. Kyrgyz Republic: Debt Sustainability Analysis[1]

Since the last debt sustainability analysis (DSA) update was published in December 2013,[2] the debt burden indicators for the Kyrgyz Republic have deteriorated. The faster-than-expected disbursement of Public Investment Related (PIP)-related loans (most financed from China), the depreciation of the som and the regional economic slowdown resulted in a significant rise of the external public debt. On balance, the Kyrgyz Republic continues to face a moderate risk of debt distress but risks have increased. A continued fall of the som, in particular, could significantly worsen the debt outlook. Stress tests suggest that the country is vulnerable to large exogenous shocks calling for the strengthening of debt management within a credible and prudent macro-fiscal framework. Fiscal consolidation and prudent use of concessional loans, in particular for large-scale public investment programs, remain key to keeping the debt outlook sustainable.

Background

The Kyrgyz Republic's nominal stock of public and publicly guaranteed (PPG) external debt, the bulk of the country public debt,[3] declined from about 100 percent of GDP in 2003 to 45 percent in 2008. This decline was mainly the result of acceleration in the pace of economic growth and a nominal appreciation, but also reflected firm fiscal discipline and Paris Club support. The Kyrgyz Republic had two debt restructuring agreements with the Paris Club, the first one in December 2002 under Houston terms and the second one in March 2005 under the Evian approach.

The nominal stock of PPG external debt significantly increased in 2009–10 to reach 56 percent of GDP in 2010. This increase resulted from the international financial crisis and domestic political instability which negatively impacted real growth but also from an uptick in external borrowing to finance infrastructure development and a crisis-related fiscal deficit.

By the end 2013, the nominal PPG external debt in percent of GDP had, however, come back to its 2008 level at 43.7 percent thanks to a strong economic recovery in 2011 and 2012, despite the launch of large public investment projects[4] to improve Kyrgyz infrastructure.

The debt outlook, however, deteriorated in 2014, with the nominal PPG external debt to GDP ratio reaching 51.0 percent. This degradation is essentially due to three factors: (i) lower growth due to the regional economic slowdown and a drop in gold production; (ii) the depreciation of the som against the dollar (following the weakening of the ruble and the tenge); and (iii) the

[1] Prepared by the staffs of the International Monetary Fund and the International Development Association.

[2] See The Kyrgyz Republic: Fifth Review Under the Three-Year Arrangement Under the Extended Credit Facility, and Request for Modification of Performance Criteria, CR 13/376

[3] The authorities committed to clarify the exact scope of the SOE's debt inclusion in the public debt by mid-March. The 10 largest SOEs will be included in the public sector.

[4] The PIP targets essentially the sectors of energy and transport and is financed by both FDI and concessional loans (see Annex I).

faster-than-expected disbursement of the Chinese part of the PIP related loans. The authorities initially budgeted total disbursements in 2014 to represent US$331 million, an amount close to 2013 disbursements, whereas they actually reached US$445 million, of which US$363 million came from EximBank China. Currently, the present value (PV) of the PPG external debt is equivalent to US$2.4 billion (36 percent of GDP), of which 44 percent is owed to international financial institutions (IFIs), and the remaining 56 percent to bilateral creditors.

Public domestic debt represented only 3.8 percent of the total public debt in 2014. This share has decreased continuously over the eight past years due to operating fiscal surplus that more than compensated for the increase in domestically financed capital expenditures.

Underlying DSA assumptions

The macroeconomic assumptions reflect the framework underlying the prospective ECF arrangement and World Bank and IMF staff projections through 2035[5].

The framework assumes the implementation of sound macroeconomic policies—including fiscal consolidation over the medium term after the deterioration of the overall balance in 2015 and prudent debt management as the basis for sustaining growth. Near medium-term growth is likely to be hindered by the regional slowdown and the uncertainties related to the accession to the Eurasian Economic Union. Over the medium term, stronger growth would be supported by the impact of the current infrastructure investments and stable gold production. Longer term growth will be underpinned by firm implementation of structural reforms to remove impediments to private investment, stimulate economic diversification, and improve the business climate, including financial stability and deepened financial intermediation. The energy sector, which has large potential, is also envisioned to add positively to growth in the long run. The framework features average medium-term GDP growth of 4.4 percent per year and average long-term growth over 2020–35 similar to the previous DSA of about 5 percent (see Text Table 1 and Box1).

Text Table 1. Kyrgyz Republic: Selected Indicators, 2013–outer years

	2013	2014	2015	2016	2017	2018	Long term
	Act.	Est.			Proj.		
Real GDP growth							
Current DSA	10.5	3.6	1.7	3.4	5.2	5.2	5.2
Previous DSA	7.8	6.5	5.3	5.0	5.0	5.0	5.0
Overal fiscal balance (percent of GDP)							
Current DSA	-3.6	-3.7	-7.6	-6.5	-3.9	-1.3	-2.5
Previous DSA	-5.2	-4.1	-2.6	-2.3	-2.5	-2.5	-2.5
Current account balance (percent of GDP)							
Current DSA	-15.0	-13.7	-17.0	-15.1	-13.4	-10.6	-7.9
Previous DSA	-10.4	-15.7	-15.1	-12.4	-9.8	-5.5	-4.3

Sources: Kyrgyz authorities and IMF staff estimates and projections.

[5] With a CPIA of 3.58, the country is assessed as a medium performer in terms of policy performance.

The main difference relative to the 2013 DSA update reflects new information regarding the authorities' infrastructure improvement plan. In 2013, only part (US$200 million) of the US$800 million Chinese PIP loans were included in the analysis. Considering the high and faster than expected execution rate of these loans in 2014,[6] the full amount of currently estimated disbursements of about US$830 million has been integrated in the DSA (see Text Table 2). These loans concern essentially two projects, the rehabilitation of a heating and power plant in Bishkek and the construction of a North-South alternative highway. More generally and contrary to the previous exercise, this DSA does not include assumptions of under execution of any of the PIP disbursements. Overall the total amount of PIP loans disbursements for the period 2015–19 is expected to represent US$1,930 million.

Text Table 2. Kyrgyz Republic: Comparison of Debt Ratio, 2013–19

	2013 Act.	2014 Est.	2015	2016	2017	2018	2019 Proj.
PPGE debt to GDP ratio							
Current DSA	43.7	51.0	57.0	59.5	59.5	56.4	53.7
Previous DSA	45.3	45.1	45.0	44.3	41.5	39.7	38.1
Public debt to GDP ratio							
Current DSA	46.1	53.0	58.8	61.2	61.0	57.7	54.9
Previous DSA	48.0	48.3	48.1	48.7	46.4	44.9	44.3

Sources: Kyrgyz authorities and IMF staff estimates and projections.

While there are many uncertainties on the economic impact of these large investments on aggregate output, empirical evidence suggests that the marginal productivity of electricity-generating capacity and paved roads is significantly higher than other infrastructure investments.[7] The efficiency of investments over the short term is probably not high given the large import content of the PIP resulting in modest GDP growth over the medium term. However, these investments have the potential to significantly contribute to longer-term growth. The energy sector has significant growth potential due to favorable export prospects and relatively low cost of power generation. However, the sector faces numerous challenges in terms of financial viability and network maintenance that the prospective increase in tariffs and the reforms in the governance of the sector should help settle. According to the authorities' feasibility study, the internal rate of return of the North-South road would significantly exceed its financing costs. Despite the strategic importance of these projects, an independent appraisal and physical monitoring of these projects to ensure their viability would be useful. More specifically, Public Investment Management (PIM) practices need to be upgraded, including by consolidating PIM responsibilities, putting in place transparent mechanisms and guidelines for screening, independent appraisal, monitoring, and evaluation.

[6] The budget 2014 included Chinese PIP loans disbursements of US$200 million whereas US$360 million were disbursed.

[7] Candelon B., Colletaz G., Hurlin C. (2013), "Network Effects and Infrastructure Productivity in Developing Countries", Oxford Bulletin of Economics and Statistics, 75(6), 887–913.

Box 1. Kyrgyz Republic: Macroeconomic Assumptions for 2015–35

Real GDP growth: After dropping in 2014 and 2015, due to the regional slowdown, inclement weather affecting the energy and agriculture sectors in 2014, and lower gold exports in 2015, real GDP is expected to gradually recover to about 5 percent in the medium-long terms. Growth in the medium term would be supported by infrastructure and energy investments, strong private sector credit growth and by new mining projects. Long term growth would be generated by firm implementation of structural reforms to remove impediments to private investment, stimulate economic diversification, and improve the business climate.

Public investment: The ratio of capital expenditure is projected to reach 10.1 percent of GDP in 2016 due to large disbursements associated with the PIP loans. It should however gradually decline over the medium term to represent about 7–8 percent of GDP.

Current account deficit: The current account deficit is expected to widen in 2015 to reach 17 percent of GDP due to lower gold exports, the import content of the energy and transport projects and the drop of remittances in the context of the regional slowdown and the ruble depreciation. Both export and import growth is expected to decelerate as the transit trade declines following the accession to the Eurasian Customs Union. The current deficit level should progressively decline over the medium term but remain relatively high during the implementation of the PIP. It should be further reduced over the long term in line with the diversification of the economy and the fiscal consolidation.

FDI: FDI is expected to rebound over the medium term after its significant decrease in 2014 and projected decline in 2015, to a level close to the historical average of 6.2 percent of GDP. After the implementation of the main public infrastructure projects, their share as a percentage of GDP should however decrease to represent about 3 percent of GDP over the long run.

Inflation: Inflation is expected to remain above 10 percent in 2015 and then gradually decrease to around 7–8 percent per annum over the medium term and to about 4 percent over the long run.

Fiscal deficit: The primary deficit is expected to widen in 2015 to reach 6.7 percent of GDP owing to increasing capital expenditures. However expenditures are expected to decline over the medium term enabling the primary deficit to stabilize at about 1.2 percent of GDP in the long run.

Financing: Over the medium term, the authorities are expected to rely on FDI and external concessional borrowing to finance the infrastructure projects. The average grant element of new external borrowing is projected to fall from 41 percent to 24 percent over the projection period, as the Kyrgyz Republic gradually moves away from concessional borrowing toward non-concessional borrowing from bilateral and commercial creditors. Meanwhile, the share of domestic financing is expected to gradually increase over the long term. Remittances are expected to remain a major source of financing but their share in percent of GDP is expected to decrease over the long term reflecting the diversification of the economy. The decline should be significant in 2015 due to the regional slowdown and more gradual over the longer term (remittances should represent around 20 percent on average on the long term). Government deposits (around 4–5 percent of GDP at end-2014) are also expected to be reduced slightly and finance the fiscal deficit.

External debt sustainability analysis

The baseline scenario points to a deterioration of the external debt burden over the medium term in comparison with the previous DSA. The present value (PV) of public and publicly guaranteed (PPG) debt to GDP and remittances ratio is now expected to reach 32.4 percent in 2016, a level much closer to the indicative threshold of 36 percent than it used to be (see Table 1 and Figure 1).[8] However, the PV of the debt to exports and remittances ratio and the PV of debt to revenue, at 62 and 133 respectively in 2016, are expected to remain well below their policy-based indicative thresholds.[9]

The debt service is expected to remain manageable throughout the DSA horizon. This reflects the high concessionality of both the outstanding multilateral debt and the assumed new borrowing over the medium term. The debt service to exports and remittances is expected to increase slightly over the projection period from 2.9 percent in 2015 to 4.4 percent in 2035. The debt service to revenue ratio would also slightly increase, from 6.2 to 8.1 percent.

The debt ratios are consistently lower in the historical scenario than the baseline over the medium term. This is because the non-interest current account deficit has been lower on average from 2005 to 2014 (reflecting a strong consolidation effort until 2008) than projected for the period 2015–20 in the baseline scenario, and the net inflows of FDI higher. Thus, applying the historical numbers leads to lower gross financing needs and a lower present value of external public debt.

Stress tests, however, show that the Kyrgyz Republic's external debt is vulnerable to large shocks or substantially less favorable assumptions. The PV of the debt to GDP and remittances ratio rises above the relevant indicative threshold over the medium term under four of the six stress tests: (i) when the exports grow at one standard deviation below their historical average over 2016–17; (ii) when the net non-debt creating inflows over 2016–17 are one standard deviation below their historical average; (iii) when the exchange rate depreciates by 30 percent in 2016 relative to the baseline; and (iv) under a shock over 2016–17 combining lower GDP and export growth, a decline in the U.S. dollar GDP deflator and lower net non-debt creating inflows (see Table 2). The shock on the exchange rate in particular would lead to a significant and protracted breach of the indicative threshold over the medium term. However, none of the other debt burden indicators breaches its indicative threshold under similar stress tests.

[8] The use of the remittances-adjusted thresholds is justified by the large size of remittances: they represented more than 30 percent of the GDP and more than 60 percent of the exports of goods and services on average over the three past years.

[9] The relevant thresholds are 120 percent for the PV of the debt to exports and remittances ratio and 250 percent for the PV of the debt to revenue ratio.

Public debt sustainability analysis

Domestic public debt is projected to increase and will play a more important role in financing the budget deficit in the medium and long term. Domestic debt accounted for about 3.8 percent of total public debt at end-2014 while it is expected to grow to 39 percent of total public debt by 2035 as domestic financial markets deepen. Currently, there is no evidence that domestic debt is crowding out private investments and the gradual increase in domestic public debt should ensure that this remains so.

The Kyrgyz Republic's total public debt outlook is projected to be manageable in the medium and long terms. Under the baseline scenario, the PV of public debt to GDP ratio would increase slightly from 38.0 percent by end-2014 to 42.2 in 2016 following disbursements for the public infrastructure projects. It would then decline over the medium term reflecting fiscal consolidation and stabilize at about 42 percent (see Table 3 and Figure 2).

Alternative scenarios and stress tests show that the Kyrgyz Republic's public debt remains highly sensitive to shocks that reduce real GDP growth or a failure to reduce the primary deficit over the medium term from the 2015 level, which result in unsustainable debt trajectories. Under stress tests and alternative scenario, the PV of the debt to GDP ratio in 2035 could increase to about 90 percent under the permanent real GDP growth shock scenario and to about 100 percent under the fixed (at 2015 level) primary deficit scenario (see Table 4). Fixing the primary deficit and the GDP growth at their historical average would also lead to a breach of the indicative threshold of 56 percent with a debt to GDP ratio reaching 65 percent in 2035.

Conclusion

While the Kyrgyz Republic's external public debt has significantly increased in 2014, the country is assessed to remain at a moderate risk of debt distress. All PV-based external indicators in the baseline are projected to stay below their indicative thresholds over the DSA horizon. Moreover, the debt service burden would remain well below the threshold reflecting the high concessionality of external debt. However, stress tests show that the external public debt indicators could significantly breach the threshold in case of a large depreciation of the som or a significant reduction of non-debt creating inflows. In the context of the regional slowdown, these scenarios are not unlikely: the som depreciated over 2014 by around 20 percent against the dollar; the remittances slowed down; and FDI decreased. This situation requires a careful management of the external public debt. The authorities broadly agreed with these results and committed to monitor it more closely.

Staff recognizes that the country has large infrastructure needs; however, safeguards are needed to keep debt at sustainable levels. It would be important to ensure that the limited fiscal space is used for viable projects ant that market risks, including exchange rate risk, are accounted for, so as to avoid the buildup of an unsustainable debt burden. The authorities should: (i) put in place a transparent mechanism and guidelines for independent appraisal, physical monitoring, and evaluation of public investments; (ii) consolidate public finances; and (iii) borrow only on

concessional terms. The set up of a medium-term debt management strategy should help the authorities to prioritize their investments to ensure public debt sustainability.

These conclusions remain valid when adding the domestic public debt to the analysis. Total public sector debt seems manageable in light of the dynamics of the domestic debt stock but its evolution is highly sensitive to a shock on GDP growth and to the path of fiscal consolidation.

Table 1 .Kyrgyz Republic: External Debt Sustainability Framework, Baseline Scenario, 2012–35 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**80.2**	**72.9**	**81.2**			**88.3**	**89.1**	**87.0**	**84.3**	**81.7**	**79.6**		**64.2**	**53.7**	
of which: public and publicly guaranteed (PPG)	46.3	43.7	51.0			57.0	59.5	59.5	56.4	53.7	52.0		42.7	31.7	
Change in external debt	3.3	-7.3	8.3			7.1	0.9	-2.1	-2.7	-2.6	-2.1		-1.9	-0.1	
Identified net debt-creating flows	6.5	-1.5	10.0			11.8	6.5	3.3	0.3	-0.1	-1.2		-0.5	1.0	
Non-interest current account deficit	**14.9**	**14.2**	**12.9**	**7.6**	**6.6**	**15.5**	**13.7**	**12.0**	**9.5**	**8.3**	**6.1**	**10.9**	**4.4**	**3.5**	4.2
Deficit in balance of goods and services	44.3	40.5	43.0			41.1	40.9	37.3	33.9	31.5	29.1		23.8	19.4	
Exports	49.0	51.2	42.5			41.1	40.5	40.5	39.4	39.4	39.5		40.9	42.5	
Imports	93.2	91.7	85.4			82.2	81.4	77.8	73.4	70.9	68.5		64.8	61.9	
Net current transfers (negative = inflow)	-31.2	-31.1	-32.4	-28.1	3.6	-28.0	-29.1	-27.3	-26.3	-25.8	-24.7	**-26.9**	-21.1	-16.9	-19.9
of which: official	-1.0	-1.7	-4.2			-3.9	-3.7	-1.1	-0.5	-0.5	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	1.9	4.8	2.3			2.5	2.0	1.9	1.9	2.6	1.8		1.7	1.0	
Net FDI (negative = inflow)	**-4.4**	**-8.5**	**-3.0**	**-6.2**	**3.0**	**-3.9**	**-5.8**	**-5.8**	**-6.2**	**-6.0**	**-5.7**	**-5.6**	**-4.2**	**-2.9**	-3.9
Endogenous debt dynamics 2/	**-4.0**	**-7.1**	**0.1**			**0.1**	**-1.4**	**-2.8**	**-3.1**	**-2.4**	**-1.6**		**-0.7**	**0.3**	
Contribution from nominal interest rate	0.7	0.8	0.8			1.5	1.4	1.4	1.1	1.6	2.4		2.3	2.9	
Contribution from real GDP growth	0.6	-7.6	-2.6			-1.3	-2.8	-4.2	-4.1	-4.0	-4.0		-3.0	-2.5	
Contribution from price and exchange rate changes	-5.4	-0.4	1.9			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.2**	**-5.8**	**-1.8**			**-4.7**	**-5.7**	**-5.4**	**-3.0**	**-2.5**	**-1.0**		**-1.4**	**-1.1**	
of which: exceptional financing	0.0	0.0	0.0			0.0	-0.4	-0.3	-0.3	-0.3	-0.3		-0.2	0.0	
PV of external debt 4/	...	...	66.2			70.3	70.2	68.2	66.7	65.2	63.9		52.1	44.6	
In percent of exports	...	...	155.8			170.8	173.4	168.3	169.0	165.4	161.9		127.2	104.9	
PV of PPG external debt	**...**	**...**	**36.0**			**39.0**	**40.6**	**40.6**	**38.8**	**37.2**	**36.3**		**30.6**	**22.6**	
In percent of exports	**...**	**...**	**84.8**			**94.8**	**100.2**	**100.3**	**98.3**	**94.3**	**92.0**		**74.7**	**53.2**	
In percent of government revenues	**...**	**...**	**110.9**			**128.9**	**133.1**	**130.9**	**120.7**	**114.6**	**111.1**		**95.0**	**70.9**	
Debt service-to-exports ratio (in percent)	**13.2**	**19.7**	**15.1**			**18.4**	**20.5**	**20.0**	**19.5**	**22.1**	**25.4**		**25.0**	**29.7**	
PPG debt service-to-exports ratio (in percent)	**4.5**	**7.7**	**3.7**			**4.5**	**5.4**	**5.4**	**5.7**	**6.0**	**5.7**		**6.2**	**6.1**	
PPG debt service-to-revenue ratio (in percent)	**6.9**	**12.3**	**4.8**			**6.2**	**7.2**	**7.1**	**7.0**	**7.2**	**6.9**		**7.9**	**8.1**	
Total gross financing need (Billions of U.S. dollars)	1.1	1.2	1.2			1.4	1.3	1.2	1.0	1.1	1.1		1.7	4.1	
Non-interest current account deficit that stabilizes debt ratio	11.6	21.5	4.6			8.4	12.9	14.1	12.3	10.9	8.3		6.3	3.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-0.9	10.5	3.6	4.1	4.0	1.7	3.4	5.2	5.2	5.2	5.3	4.3	4.9	5.1	5.2
GDP deflator in US dollar terms (change in percent)	7.5	0.4	-2.6	9.1	12.1	-2.1	3.8	4.0	3.3	2.7	2.0	2.3	2.0	2.0	2.0
Effective interest rate (percent) 5/	1.0	1.2	1.1	1.1	0.3	1.8	1.7	1.7	1.3	2.1	3.1	2.0	3.7	5.7	4.4
Growth of exports of G&S (US dollar terms, in percent)	3.4	16.0	-16.3	13.5	20.3	-3.6	5.7	9.4	5.8	7.9	7.5	5.4	7.2	7.8	7.8
Growth of imports of G&S (US dollar terms, in percent)	25.7	9.2	-5.9	21.4	25.4	-4.2	6.3	4.6	2.4	4.4	3.7	2.9	6.3	6.8	6.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	41.2	40.8	42.2	44.2	42.6	42.8	42.3	36.3	23.8	32.0
Government revenues (excluding grants, in percent of GDP)	32.4	32.0	32.5			30.2	30.5	31.0	32.1	32.4	32.7		32.2	31.9	32.0
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.3			0.7	0.6	0.6	0.4	0.4	0.4		0.4	0.6	
of which: Grants	0.2	0.2	0.3			0.3	0.3	0.3	0.2	0.2	0.1		0.2	0.2	
of which: Concessional loans	0.1	0.1	0.1			0.3	0.3	0.3	0.2	0.2	0.3		0.3	0.3	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			8.4	6.8	5.7	3.4	3.1	2.6		2.1	1.6	1.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			61.6	57.6	59.9	63.4	63.4	59.0		52.6	36.7	47.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	6.6	7.3	7.4			7.4	7.9	8.7	9.4	10.1	10.9		15.8	31.1	
Nominal dollar GDP growth	6.6	11.0	0.9			-0.5	7.4	9.4	8.6	8.0	7.3	6.7	6.9	7.1	7.2
PV of PPG external debt (in Billions of US dollars)			2.4			2.8	3.2	3.5	3.6	3.7	3.9		4.8	7.0	
(PVt-PVt-1)/GDPt-1 (in percent)						**5.1**	4.9	3.8	1.7	1.3	1.8	3.1	1.1	1.4	1.1
Gross workers' remittances (Billions of US dollars)	2.0	2.2	2.1			1.8	2.0	2.3	2.4	2.6	2.7		3.3	5.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	28.1			31.4	32.4	32.2	30.8	29.6	29.1		25.3	19.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	51.0			59.8	61.7	60.9	59.5	57.4	56.5		49.3	38.1	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	2.2			2.9	3.3	3.3	3.5	3.6	3.5		4.1	4.4	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

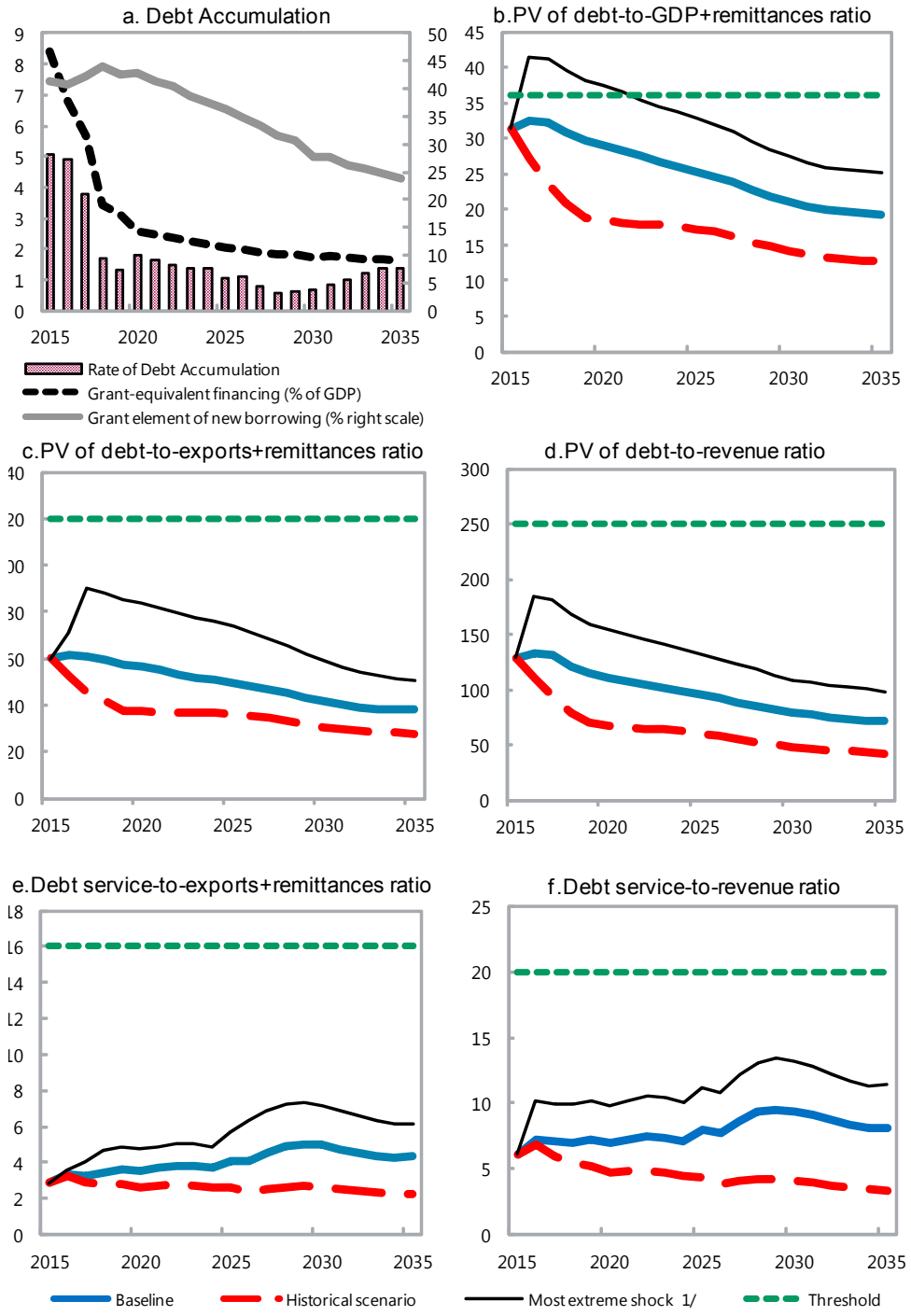
5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Figure 1. Kyrgyz Republic: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2015–35 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Table 2. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35 (continued)

(In percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to-GDP+remittances ratio								
Baseline	31	32	32	30.8	29.6	29.1	**25.3**	19
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	31	27	24	21	19	18	**17**	13
A2. New public sector loans on less favorable terms in 2015-2035 2	31	34	34	34	33	33	**31**	32
A3. Alternative Scenario : One time 10 percent drop in remittances	31	32	31	29	27	27	22	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	31	33	34	33	31	31	**27**	20
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	31	35	40	39	37	37	**32**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	31	34	35	34	33	32	**28**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	31	37	40	37	36	35	**30**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	31	36	41	39	37	37	**32**	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	31	41	41	40	38	37	**33**	25
PV of debt-to-exports+remittances ratio								
Baseline	60	62	61	59	57	57	**49**	38
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	60	52	46	41	38	37	**36**	28
A2. New public sector loans on less favorable terms in 2015-2035 2	60	64	65	65	64	64	**61**	62
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	60	61	60	59	57	56	**49**	37
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	60	71	90	88	85	84	**74**	50
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	60	61	60	59	57	56	**49**	37
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	60	73	77	72	70	69	**59**	41
B5. Combination of B1-B4 using one-half standard deviation shocks	60	70	78	74	71	70	**61**	42
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	60	61	60	59	57	56	**49**	37
PV of debt-to-revenue ratio								
Baseline	129	133	131	121	115	111	**95**	71
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	129	110	95	80	70	67	**60**	42
A2. New public sector loans on less favorable terms in 2015-2035 2	129	138	140	133	128	126	**118**	116
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	129	135	140	130	123	119	**102**	76
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	129	143	164	152	144	140	**119**	77
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	129	140	148	137	130	126	**108**	80
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	129	149	158	147	139	135	**114**	76
B5. Combination of B1-B4 using one-half standard deviation shocks	129	147	168	155	148	143	**122**	80
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	129	184	181	168	159	154	**132**	98

Table 2. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35 (concluded)

(In percent)

Debt service-to-exports+remittances ratio								
Baseline	2.9	3.3	3.3	3.5	3.6	3.5	**4.1**	4.4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	3	3	3	3	3	3	**3**	2
A2. New public sector loans on less favorable terms in 2015-2035 2	3	3	3	4	4	4	**4**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	3	3	3	3	4	4	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	3	4	4	5	5	5	**6**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	3	3	3	3	4	4	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	3	4	4	4	4	4	**5**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	4	4	4	4	**5**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	3	3	3	3	4	4	**4**	4
Debt service-to-revenue ratio								
Baseline	6.2	7.2	7.1	7.0	7.2	6.9	**7.9**	8.1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	6	7	6	5	5	5	**4**	3
A2. New public sector loans on less favorable terms in 2015-2035 2	6	7	7	7	8	8	**8**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	6	7	8	8	8	8	**9**	9
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	6	7	7	8	8	8	**9**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	6	8	8	8	8	8	**9**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	6	7	8	8	8	8	**9**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	6	7	8	8	9	8	**10**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	6	10	10	10	10	10	**11**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	25	25	25	25	25	25	**25**	25

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Kyrgyz Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2012–35

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate					Projections				
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	49.0	46.1	53.0			58.8	61.1	60.9	57.7	54.9	53.4		53.6	51.4	
of which: foreign-currency denominated	46.9	43.7	51.0			57.0	59.5	59.5	56.4	53.7	52.0		42.7	31.7	
Change in public sector debt	-0.4	-2.9	6.9			5.8	2.3	-0.2	-3.2	-2.8	-1.4		0.0	-0.3	
Identified debt-creating flows	2.3	-1.0	6.6			4.2	1.5	-1.7	-4.1	-4.6	-3.4		-1.3	-1.1	
Primary deficit	4.4	2.7	2.9	2.7	2.0	6.7	5.7	3.1	0.6	-0.8	0.0	2.6	1.5	1.3	1.4
Revenue and grants	34.9	34.4	36.0			35.0	33.8	33.9	34.0	34.2	33.9		33.2	32.6	
of which: grants	2.5	2.4	3.5			4.7	3.3	2.9	1.9	1.8	1.2		1.0	0.7	
Primary (noninterest) expenditure	39.3	37.2	38.9			41.6	39.6	37.1	34.6	33.4	33.9		34.7	33.9	
Automatic debt dynamics	-2.1	-3.8	3.8			-2.4	-3.8	-4.5	-4.4	-3.6	-3.1		-2.6	-2.4	
Contribution from interest rate/growth differential	-0.3	-5.2	-2.0			-0.9	-2.3	-3.4	-3.4	-3.1	-3.0		-2.6	-2.4	
of which: contribution from average real interest rate	-0.7	-0.5	-0.4			0.0	-0.4	-0.4	-0.4	-0.3	-0.3		-0.1	0.1	
of which: contribution from real GDP growth	0.4	-4.7	-1.6			-0.9	-1.9	-3.0	-3.0	-2.8	-2.7		-2.5	-2.5	
Contribution from real exchange rate depreciation	-1.8	1.4	5.7			-1.6	-1.5	-1.1	-1.0	-0.4	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	-0.4	-0.3	-0.3	-0.3	-0.3		-0.2	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	-0.4	-0.3	-0.3	-0.3	-0.3		-0.2	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.7	-1.8	0.3			1.5	0.8	1.5	0.9	1.8	1.9		1.3	0.8	
Other Sustainability Indicators															
PV of public sector debt	...	...	38.0			40.8	42.2	42.1	40.1	38.3	37.8		41.5	42.3	
of which: foreign-currency denominated	...	...	36.0			39.0	40.6	40.6	38.8	37.2	36.3		30.6	22.6	
of which: external	...	...	36.0			39.0	40.6	40.6	38.8	37.2	36.3		30.6	22.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	8.2	7.7	6.1			9.7	8.9	6.2	3.6	2.2	3.1		10.0	6.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	105.6			116.7	124.8	124.0	117.9	112.2	111.3		124.8	129.5	
PV of public sector debt-to-revenue ratio (in percent)	...	...	117.2			134.9	138.4	135.6	124.7	118.3	115.5		128.7	132.6	
of which: external 3/	...	...	110.9			128.9	133.1	130.9	120.7	114.6	111.1		95.0	70.9	
Debt service-to-revenue and grants ratio (in percent) 4/	6.5	11.6	5.6			6.1	7.0	6.9	6.9	7.1	7.6		14.3	18.6	
Debt service-to-revenue ratio (in percent) 4/	7.0	12.5	6.2			7.0	7.8	7.5	7.3	7.5	7.8		14.8	19.0	
Primary deficit that stabilizes the debt-to-GDP ratio	4.8	5.6	-4.1			0.9	3.4	3.3	3.8	2.1	1.4		1.5	1.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-0.9	10.5	3.6	4.1	4.0	1.7	3.4	5.2	5.2	5.2	5.3	4.3	4.9	5.1	5.2
Average nominal interest rate on forex debt (in percent)	1.2	1.2	1.3	1.1	0.2	1.4	1.1	1.1	1.1	1.3	1.3	1.2	1.6	2.3	1.8
Average real interest rate on domestic debt (in percent)	5.5	16.7	6.0	1.6	8.1	3.7	2.4	1.9	1.9	1.9	0.0	2.0	0.0	0.0	0.0
Real exchange rate depreciation (in percent, + indicates depreciation	-3.9	3.4	13.8	-3.9	10.7	-3.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	9.5	3.5	7.9	11.2	6.9	10.1	8.0	7.1	5.9	4.7	4.0	6.6	4.0	4.0	4.0
Growth of real primary spending (deflated by GDP deflator, in percer	6.3	4.6	8.4	2.0	3.2	8.8	-1.7	-1.5	-1.7	1.5	6.8	2.0	4.5	6.0	5.2
Grant element of new external borrowing (in percent)	...	...	...	...	...	41.2	40.8	42.2	44.2	42.6	42.8	42.3	36.3	23.8	...

Sources: Country authorities; and staff estimates and projections.

1/ General government gross debt.

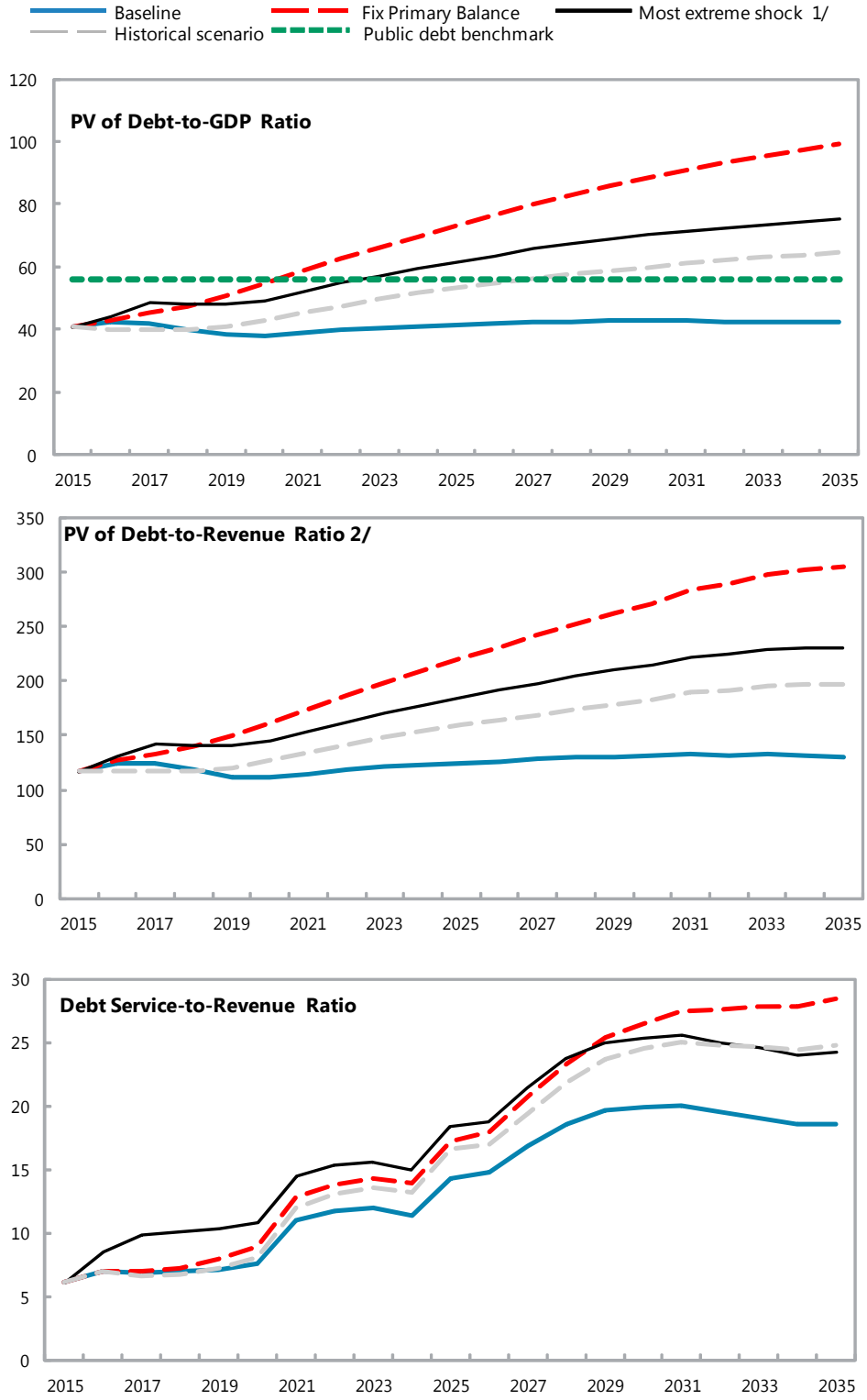
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Figure 2. Kyrgyz Republic: Indicators of Public Debt Under Alternative Scenarios, 2015–35 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Table 4. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public Debt 2015–35

	Projections							
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	41	42	42	40	38	38	41	42
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	41	40	40	40	41	43	53	65
A2. Primary balance is unchanged from 2015	41	43	45	48	51	55	73	99
A3. Permanently lower GDP growth 1/	41	43	43	42	42	42	55	88
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	41	44	48	48	48	49	62	75.3
B2. Primary balance is at historical average minus one standard deviations in 2016-201	41	42	43	41	39	38	42	43
B3. Combination of B1-B2 using one half standard deviation shocks	41	41	43	42	41	41	50	58
B4. One-time 30 percent real depreciation in 2016	41	58	56	52	50	49	50	49
B5. 10 percent of GDP increase in other debt-creating flows in 2016	41	50	49	47	45	44	47	46
PV of Debt-to-Revenue Ratio 2/								
Baseline	117	125	124	118	112	111	125	130
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	117	118	117	117	120	126	160	197
A2. Primary balance is unchanged from 2015	117	127	134	140	149	162	221	305
A3. Permanently lower GDP growth 1/	117	126	128	124	121	124	166	268
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	117	131	141	141	140	145	185	230
B2. Primary balance is at historical average minus one standard deviations in 2016-201	117	123	126	119	114	113	126	130
B3. Combination of B1-B2 using one half standard deviation shocks	117	122	126	123	120	122	150	177
B4. One-time 30 percent real depreciation in 2016	117	170	164	154	146	143	151	152
B5. 10 percent of GDP increase in other debt-creating flows in 2016	117	146	145	137	131	130	141	140
Debt Service-to-Revenue Ratio 2/								
Baseline	6	7	7	7	7	8	14	19
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	6	7	7	7	7	8	17	25
A2. Primary balance is unchanged from 2015	6	7	7	7	8	9	17	28
A3. Permanently lower GDP growth 1/	6	7	7	7	7	8	16	27
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	6	7	7	8	8	9	17	25
B2. Primary balance is at historical average minus one standard deviations in 2016-201	6	7	7	7	7	8	14	19
B3. Combination of B1-B2 using one half standard deviation shocks	6	7	7	7	7	8	16	22
B4. One-time 30 percent real depreciation in 2016	6	8	10	10	10	11	18	24
B5. 10 percent of GDP increase in other debt-creating flows in 2016	6	7	8	8	8	8	15	20

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.